PYL703 Quiz 2 reconstructed follow up prompts

These ten prompts were drafted during the review. They are suggestions for a step by step recheck, not a transcript of earlier student messages. Use the actual chat export for the assignment history requirement.

Step 1

read the quiz pdf and spreadsheet first. list what we need to answer, and check the submission limits before doing the physics.

Step 2

recalc f = abs(theta)/torder for all 22 entries. keep missing values blank, and flag anything that doesnt match the sheet.

Step 3

which theta signs disagree with the listed order? separate actual competing interactions from canting, uncertain fits, and possible data errors.

Step 4

check mnbi2te4 in its paper. how can fm layers still order antiferromagnetically, and is the reported positive theta actually significant?

Step 5

check gdmno3's 2 k value and the chromite sign columns against the papers. dont silently replace the database values with another sample's numbers.

Step 6

compare hocro3 and ercro3 with ycro3 and lucro3 using the sheet. does small abs(theta) really mean weak cr exchange?

Step 7

the hexagonal manganites have f around 4 to 6. explain the frustration argument, but check whether that tells us they lack magnetic order.

Step 8

open the hexagonal ymno3 paper and verify -417 k and 71 k. also check the fit range and fields; being above tn may not be enough.

Step 9

pick one case that follows the simple sign-and-scale picture and one that needs extra evidence. be clear about what two temperatures can actually establish.

Step 10

do a final audit for unsupported claims and citation mixups. include a real ai mistake from the saved chat, then keep the answer within two pages.